

AI Recipe Generator – Product Requirements Document (PRD)

1. Project Overview

Project Name:

AI Recipe Generator

Platform:

Flutter (Android First)

Purpose:

Develop a modern AI-powered recipe application that allows users to discover, generate, and save recipes using AI (Gemini). The app should have a premium UI inspired by Dribbble designs while maintaining its own branding.

2. Design Vision

The application should feel like a premium cooking assistant.

Design inspiration:

- Warm Brown Theme
- Soft Cream Background
- Large Rounded Corners
- Soft Shadows
- Minimal Interface
- Elegant Typography
- Smooth Animations
- Modern Card Layouts

Do NOT copy Dribbble directly.

Use it only as inspiration.

The UI should look original but at the same quality level.

3. Color Palette

Primary Brown

#8B5E3C

Background

#F6F2EE

Card

White

Text

#2E2E2E

Secondary Text

Grey

Accent

Soft Orange/Brown

4. Logo

Already designed.

Location:

assets/images/logo.png

The logo should appear on:

- Splash Screen

- Login
- Register
- Drawer
- Profile

5. Current Progress

Completed

Splash Screen

Completed

Login Screen

Mostly completed.

Current UI includes:

- Logo
- Welcome Heading
- Subtitle
- Email TextFormField
- Password TextFormField
- Floating Labels
- Password Visibility Toggle
- Remember Me
- Forgot Password
- Google Sign In Button
- Sign Up Link
- Brown Theme

Current Design Quality:

Approximately 90%

Needs only polishing.

6. Login Improvements Remaining

Improve spacing.

Improve alignment.

Polish floating labels.

Improve responsive layout.

Make keyboard interaction smoother.

Use proper form validation.

Add loading animation while signing in.

7. Registration Screen

Create similar to Login.

Fields:

- Name
- Email
- Password
- Confirm Password

Features:

Password visibility

Validation

Google Sign Up

Already have account button

8. Forgot Password

Simple page.

Email input.

Send reset email.

Success message.

9. Home Screen

This is the main focus.

Inspired by the provided Dribbble design.

Sections:

Greeting

Search Bar

AI Assistant Button

Recipe Categories

Popular Recipes

Trending Recipes

Continue Cooking

Featured Chef

Daily Recommendation

Bottom Navigation

Everything should be responsive.

10. Bottom Navigation

Five Tabs

Home

Favorites

AI Chat

Saved Recipes

Profile

The AI tab should be highlighted.

11. AI Recipe Generator

Core Feature.

User enters:

Ingredients

or

Recipe Prompt

Example:

Make a healthy chicken pasta under 500 calories.

Send prompt to Gemini API.

Display:

Recipe Name

Ingredients

Instructions

Cooking Time

Calories

Serving Size

Tips

Save Button

Share Button

12. AI Chat Assistant

Chat interface similar to ChatGPT.

Supports:

Recipe questions

Cooking techniques

Ingredient substitutions

Meal planning

Food tips

Chat bubbles.

Typing animation.

13. Search

Search recipes.

Search ingredients.

Recent searches.

Popular searches.

14. Categories

Examples:

Breakfast

Lunch

Dinner

Desserts

Snacks

Healthy

Vegan

Drinks

15. Recipe Detail

Large image.

Title.

Cooking Time.

Calories.

Difficulty.

Ingredients.

Instructions.

Nutrition.

AI Tips.

Save.

Share.

Favorite.

16. Favorites

Saved recipes.

Search.

Filter.

Delete.

17. User Profile

Profile image.

Name.

Email.

Saved recipes count.

Generated recipes count.

Settings.

Logout.

18. Firebase

Use Firebase.

Authentication

Firestore

Storage

19. Authentication

Email Password

Google Sign In

Remember Login

Forgot Password

20. Firestore Collections

Users

Recipes

Favorites

Generated Recipes

Chat History

21. Architecture

Use clean folder structure.

Example:

`lib/`

`screens/`

`widgets/`

`models/`

`services/`

`providers/`

`utils/`

`theme/`

`constants/`

22. State Management

Use Provider.

Do NOT use `setState` for business logic.

23. Code Style

Reusable widgets.

Proper comments.

Meaningful naming.

Clean architecture.

No duplicate code.

24. Animations

Fade

Slide

Hero

Page transitions

Button press

Loading

Typing

25. Responsiveness

Must work on:

Small phones

Medium phones

Large phones

Tablets

No overflow.

No clipped widgets.

26. Assets

Use copyright-free images.

Use high-quality food photos.

Icons should be modern.

27. Performance

Lazy loading.

Image caching.

Fast startup.

Smooth scrolling.

28. Accessibility

Readable fonts.

Good contrast.

Proper touch targets.

29. Future Features

Voice Recipe Search

Recipe Scanner

Barcode Scanner

AI Meal Planner

Shopping List

Dark Mode

Notifications

Weekly Meal Plans

Recipe Sharing

31. Important Instruction for Claude

Do not redesign the application with a different style.

Keep the existing warm brown theme and current branding.

Preserve the existing logo and color palette.

Every new screen should follow the same premium design language established in the Login screen and inspired by the provided Dribbble reference.

Focus on clean architecture, reusable widgets, responsiveness, and production-ready Flutter code.